



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
02-01-2022

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S(TEPS)]

Time : 3 Hours

PAPER-2

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

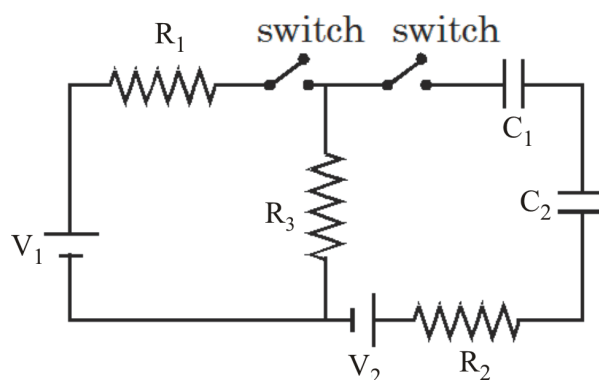
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. In the given circuit $C_1 = C$, $C_2 = 2C$, $V_1 = V_2 = V$ & $R_1 = R_2 = R_3 = R$ and all switches-are closed at $t = 0$ then choose the incorrect statement at time $t = RC$ (initially all capacitors are uncharged)



- (A) Current through V_1 is less than $\frac{V}{2Re} [e - 1]$
- (B) Power delivered by V_2 is less than $\frac{V^2}{3eR}$
- (C) Current through V_1 is greater than $\frac{V}{2Re} [e - 1]$
- (D) Power delivered by V_2 is greater than $\frac{V^2}{3eR}$

2. A conducting spherical shell of inner radius a and outer radius b is made up of a material of resistivity ρ and negligible dielectric activity. At time $t = 0$, material of shell is electrically neutral and a charge q_0 is some how located at the centre of shell (i.e. at time $t = 0$, there are no induced charges, they will appear afterward). Choose correct statement(s) :

(A) When $-Q_t$ charge is induced at inner surface at time t , then electric field in the material of shell

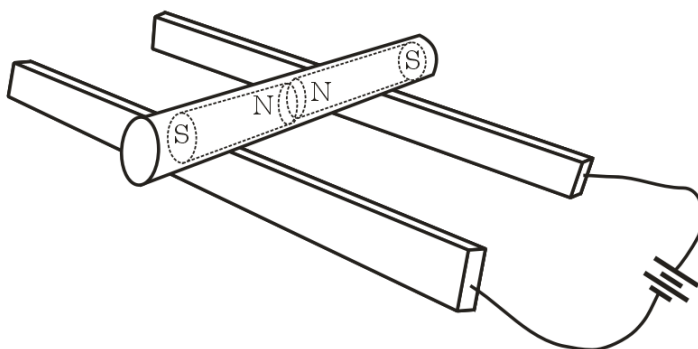
is $E = \frac{q_0 - Q_t}{4\pi\epsilon_0 r^2}$ where r is distance of point under consideration from centre of shell.

(B) Current through shell at this moment (discussed in option A) is $I = \frac{q_0 - Q_t}{\epsilon_0 \rho}$

(C) At time t we can write $\frac{dQ_t}{q_0 - Q_t} = \frac{dt}{\epsilon_0 \rho}$

(D) At time t , value of Q_t is given by $Q_t = q_0(1 - e^{-\frac{t}{\epsilon_0 \rho}})$

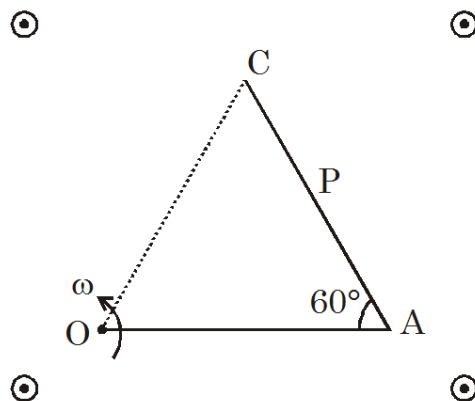
3. In the figure shown, a hollow copper tube is fitted with two cylindrical bar magnets of same magnetic dipole moment but opposite polarity. The bar magnets are fixed inside the tube so that they rotate with the tube. The south pole is outside the region of the current flow in the copper tube. Assume that the current flow in the copper tube is always parallel to its axis. When the current is switched on,



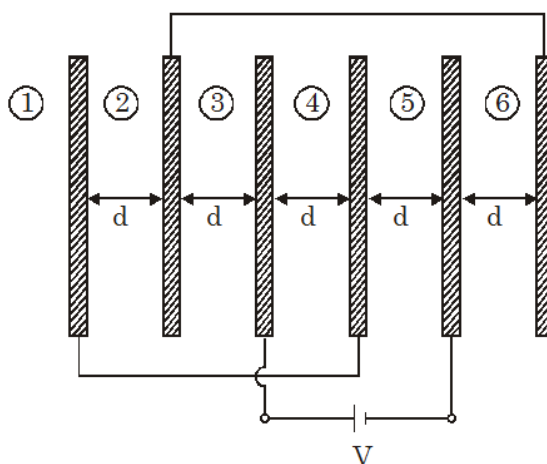
- (A) If the tube is on smooth rails, it rotates about its axis in anticlockwise direction.
 (B) If the tube is on smooth rails, it rotates about its axis in clockwise direction.
 (C) If the tube is on rough rails so that it does not slip, it rolls away from the battery.
 (D) When tube is on smooth rails, it rotates without any external torque.

Space for Rough Work

4. A metallic V shaped rod OAC is rotated with respect to one of its end in uniform magnetic field, such that axis of rotation is parallel to the direction of magnetic field. Length of each arm of rod is L and angle between the arms is 60° . P is the mid-point of section AC. Magnitude of magnetic field is B . Then choose the correct relations.



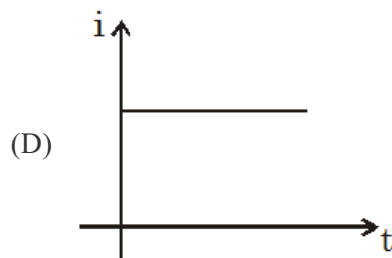
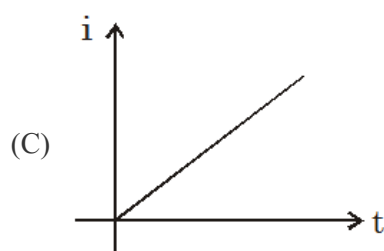
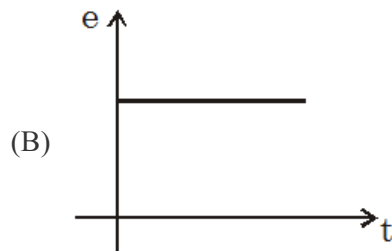
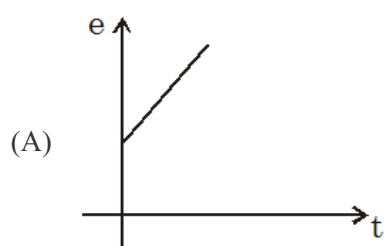
- (A) $V_A - V_O = \frac{\omega BL^2}{2}$ (B) $V_A - V_C = \frac{\omega BL^2}{2}$ (C) $V_C - V_P = \frac{\omega BL^2}{8}$ (D) $V_A - V_P = \frac{\omega BL^2}{8}$
5. Consider an arrangement of six large parallel plates as shown in the figure, each of plate has an area A . Initially when battery was not connected all plates were uncharged. The emf of the battery is V . Choose the correct option(s) after connecting the battery (in steady state):



- (A) Total work done by battery is $\frac{\epsilon_0 A}{d} V^2$.
- (B) Electric field between plates 3 and 4 is $\frac{V}{2d}$.
- (C) Potential difference between plate 2 and 5 is $\frac{V}{2}$
- (D) Potential difference between plate 1 and 5 is $\frac{V}{2}$

Space for Rough Work

6. Two parallel long straight conductors lie on a smooth horizontal surface. Two other parallel conductors rest on them at right angles so as to form a square of side a initially and all the conductors are made of same material. A uniform magnetic field B exists in vertical direction. Now all the four conductors start moving outwards with a constant velocity v . The induced e.m.f. e in loop and induced current i will vary with time t as :



Space for Rough Work

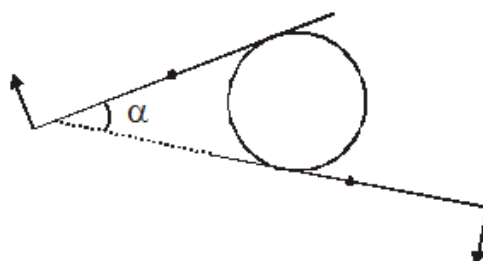
SECTION-II : (Maximum Marks: 24)

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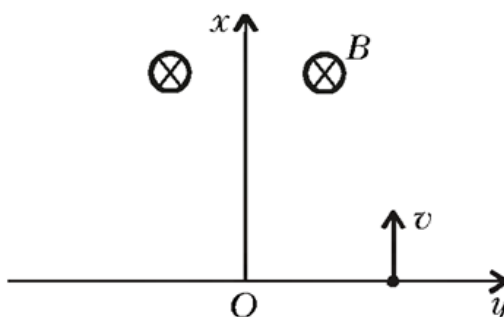
Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. A Light cylinder is sandwiched between two identical levers so that the angle between them is equal to $\alpha = 60^\circ$ (see figure). The dots denote the axis of rotation of the levers, and arrows show the forces applied to the ends of the levers. At what minimum coefficient of friction between the levers and the cylinder it may be in equilibrium in this situation? Gravity is neglected.

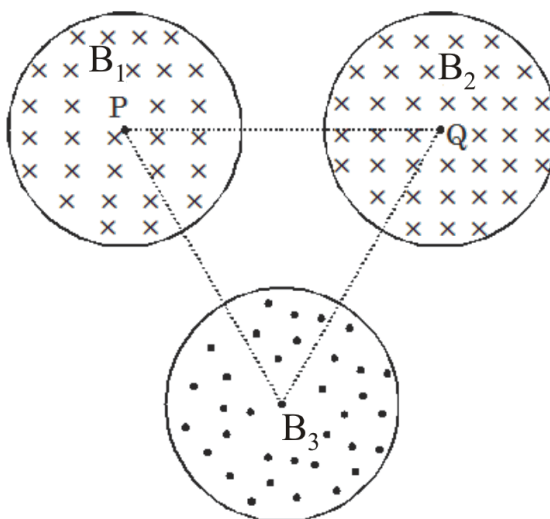


2. The current in a conductor of 60 ohm resistance increases linearly from 5 A to 25 A in 10 s. How much heat (in kJ) is liberated in it during this time ?
3. In a non-uniform magnetic field $B = ax\hat{k}$ ($x \geq 0$) a charge particle starts with mass m and charge q with initial velocity v directed along the x axis. Determine the maximum displacement (in m) of particle in $+x$ direction. Take $m = 2$ gm, $q = 100 \mu\text{C}$, $a = 0.4$ T/m, $v = 4$ m/s.



Space for Rough Work

4. In the figure B_1 , B_2 and B_3 represent uniform time varying magnetic fields in three different infinitely long cylindrical regions with axis passing through P, Q and R and perpendicular to the plane of the paper. PQR is an equilateral triangle of side $4\sqrt{3}r$, where r is the radius of each cylindrical region ($r = 1\text{m}$). The magnetic inductions vary with time t as $B_1 = A + at$, $B_2 = A - bt$ and $B_3 = A + ct$. Here $a = 2\text{ Tesla/s}$, $b = 2\text{ Tesla/s}$ and $c = 1\text{ Tesla/s}$ and A is a constant with dimensions of magnetic field and t is time. Find the induced electric field strength (in V/m) at the centroid of the triangle PQR



5. A coil of radius $r = 4\text{ cm}$ and resistance $= 0.01\text{ ohms}$ is in a uniform magnetic field $B = 10^{-2}\text{ T}$. The plane of the coil makes an angle of 30° with the magnetic field lines. If the magnetic field uniformly decreases to zero, what is the charge (in mC) flowing through the coil?
6. The flywheel of an engine begins to rotate starting from rest with constant angular acceleration of 0.5 rad/s^2 and after $t = 15\text{ s}$ after the beginning of the movement its angular momentum is $73.5\text{ kg m}^2/\text{sec}$. Find the kinetic energy (in J) of the wheel 20 s after the beginning of the rotation.

Space for Rough Work

SECTION-III : (Maximum Marks: 18)

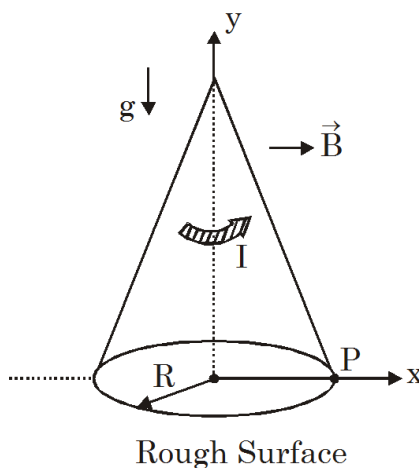
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Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

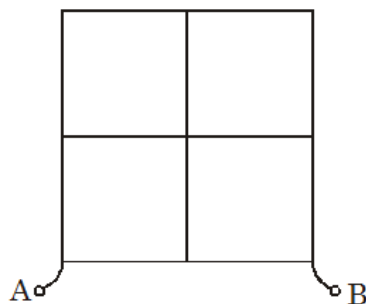
1. A current I is flowing uniformly along the curved surface of a solid cone. This cone is kept on a sufficiently rough surface in a magnetic field $\vec{B} = B_0 \hat{i}$. If the maximum value of current such that the cone doesn't topple about point P is $\frac{kmg}{\pi B_0 R}$, find the value of k ? Mass of the cone is m and its base radius is given by R .



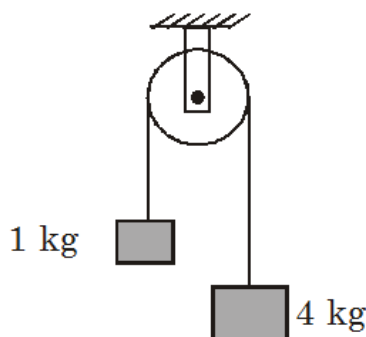
2. Inside the plane capacitor with an area $S = 200 \pi \text{ cm}^2$ and the distance between them $d = 0.01 \text{ cm}$ is a glass plate, completely filling the space between the plates of the capacitor. What work (in mJ) should be done to slowly remove the glass plate? Capacitor is permanently connected to the battery with electromotive force $E = 300 \text{ V}$. The dielectric constant of glass $K = 5$. Force of friction and the mass of the glass plate can be neglected.

Space for Rough Work

3. The figure shown below shows a circuit for which the resistance of each branch is 4Ω . Determine the electrical resistance (in ohms) between points A and B.



4. By means of a rope whose mass is negligible, two blocks are suspended over a pulley, as the drawing shows. The pulley can be treated as a uniform solid cylindrical disk. The downward acceleration of the 4 kg block is observed to be exactly one-half the acceleration due to gravity. Find the mass (in kg) of the pulley.



5. Two thin rectangular sheets ($0.20\text{m} \times 0.40\text{m}$) are identical. In the first sheet the axis of rotation lies along the 0.20m side, and in the second it lies along the 0.40m side. The same torque is applied to each sheet. The first sheet, starting from rest, reaches a particular angular velocity in 4 sec. How long (in sec) does it take for the second sheet, starting from rest, to reach the same angular velocity?
6. A conducting wire frame has an area of 10 cm^2 and is kept perpendicular to magnetic field lines. The induction of the magnetic field changes as follows: $B = B_0(a - ct^2)$ till it becomes zero. where $B_0 = 0.01\text{ T}$, $a = 9$, $c = 1/\text{sec}^2$, t is time. Resistance frame $R = 10^{-2}\text{ ohms}$. Find the maximum current (in mA) through the wire frame.

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I : (Maximum Marks: 24)

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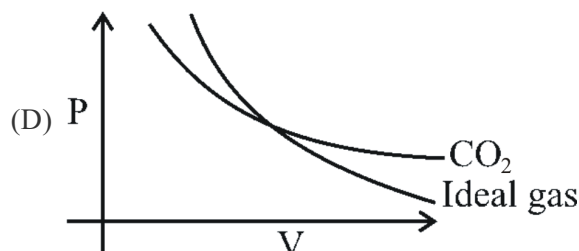
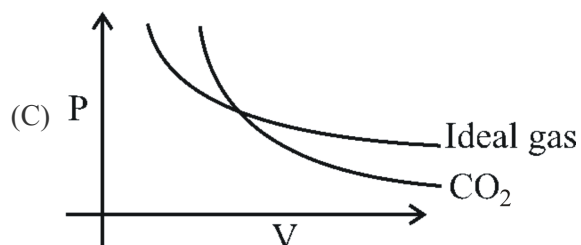
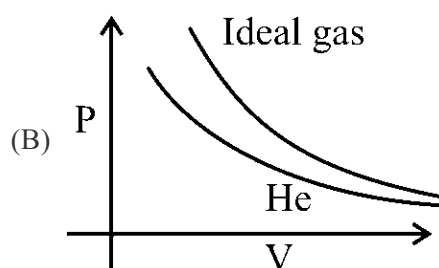
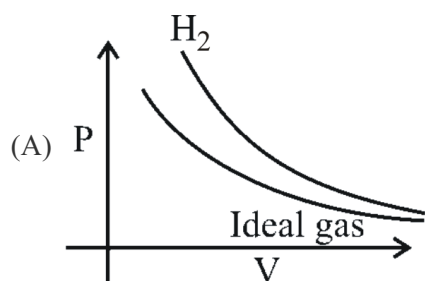
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

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- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Which of the following is/are correct near room temperature-



Space for Rough Work

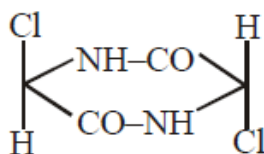
2. Identify the correct statement(s) among the following

- (A) 0.5 gm-molecule of oxygen contain $\frac{N_A}{4}$ oxygen molecules
 (B) Total electrons present in 0.5 g-atom of oxygen is $4N_A$
 (C) Mass of 3.01×10^{23} molecules of ammonia is 8.5 g
 (D) Number of atoms present in 11.2 litres of chlorine at 1 atm & 273K is 6.022×10^{23}

3. Compounds which are more reactive than ethene towards electrophilic addition of H-Cl

- (A) $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$ (B) $\text{CH}_2=\text{CH}-\text{NO}_2$
 (C) $\text{CH}\equiv\text{CH}$ (D) $\text{CH}_2=\text{CH}-\text{OMe}$

4. Which of the following statements are correct about the given compound



- (A) It has center of symmetry (B) It has plane of symmetry
 (C) It is optically inactive (D) It can rotate plane polarised light

Space for Rough Work

5. Element "X" from second period has more "IE" than element "Y" from same period. "X" forms a series of compounds with "Y" such as X_2Y_3 , X_2Y_4 , X_2Y_5 , XY , XY_2 etc.
Y and X both have least electron affinity in their respective groups.
Which is/are correct regarding "X" and "Y"?
- (A) XY_3^- has X–Y bond order equal to S–O bond order in SO_3^{2-}
 (B) $H_2X_2Y_2$ contains linkage between similar atoms
 (C) XY & X_2Y has no effect on addition of conc. HCl or dil. HCl
 (D) XY_2 is paramagnetic
6. The specie(s) that has/have maximum value of $\frac{\sigma - \text{bond}}{\text{Lone pairs}}$ among the following is/are.
- (A) SO_2 (B) SO_3 (C) XeO_4 (D) ClO_4^-

Space for Rough Work

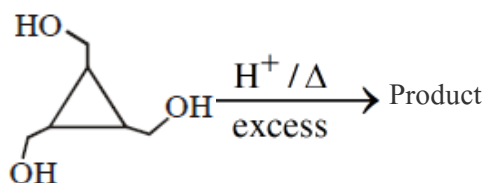
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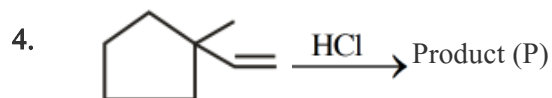
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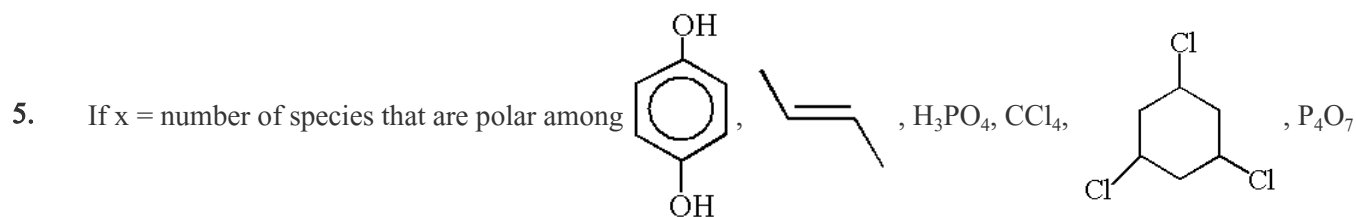
- Two elements x and y form two types of non-electrolyte molecules like xy_3 and xy . 1 gm of xy_3 lowers the freezing point by 1°C when dissolved in 100 gram solvent whereas for the same lowering in freezing point for same amount of solvent 0.5 gram xy is required. The sum of the atomic masses of x and y is Z. Find $\left(\frac{Z}{5}\right)$. (Given : k_f for solvent is 5 K molal^{-1}).
- A solution containing 2.54 gram of a salt $K_xH_y(C_2O_4)_z \cdot nH_2O$ per litre. 10 ml of this salt solution required 30 ml of 0.01 N NaOH for complete neutralisation. Same quantity of salt solution was also found to require 40 ml of 0.01 N $KMnO_4$ solution for complete oxidation. Find out x, y, z and n (x, y, z are simplest possible integers) ($K = 39$). Report your answer as $\left\{ \frac{x + y - z}{n} \right\}$
- Total number of carbon atoms in the ring frame of the product is :



Space for Rough Work

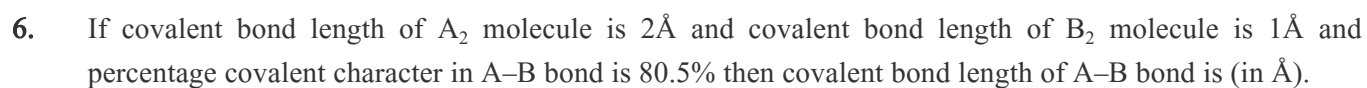


Number of stereoisomers possible for product (P) formed?



y = Maximum number of electrons that have identical value of effective nuclear charge in "Cu".

The value of $\frac{y}{x}$ is _____.



Space for Rough Work

SECTION-III : (Maximum Marks: 18)

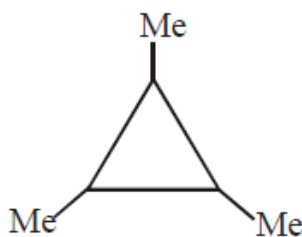
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1. How many times volume of monoatomic gas should be expanded reversibly & adiabatically in order to reduce its r.m.s velocity to half.
2. The activation energy of a second order reaction : $A + B \rightarrow P$ is $9.2 \text{ kcal mole}^{-1}$. At 500K, product is formed at the rate of $0.08 \text{ mol L}^{-1}\text{min}^{-1}$. Find the rate of formation of product at 1000 K in $\text{mol L}^{-1}\text{min}^{-1}$. if other factors are assumed to be same [Use $\ln x = 2.3 \log x$]
3. Identify total number of stereo isomers for following compound which are optically inactive.



Space for Rough Work

4. Total number of cyclic structure isomers possible with MF C_5H_{10} :
5. Number of specie(s) that share atleast one oxygen atom per tetrahedral unit among the following.
 P_4O_{10} , $[Na_2Fe_3Fe_2Si_8O_{22}(OH)_2]$, $Ca_5Si_5O_{15}$, $H_2S_2O_7$, $Si_2O_7^{6-}$, $K_2Cr_2O_7$, Borax
6. The number of degenerate orbitals in absence of external magnetic field & electric field in third excitation state of Be^{3+} is "x" times the degeneracy value in 1st excited state of He^+ . "x" is _____.
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Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I : (Maximum Marks: 24)

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- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

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1. The number of selection of four letters taken from the word COLLEGE must be
 (A) 22
 (B) 18
 (C) coefficient of x^4 in the expansion of $(1 + x + x^2)^2(1 + x)^3$
 (D) 32
 2. Let a, b, c be three distinct real numbers such that each of the expressions $ax^2 + bx + c$, $bx^2 + cx + a$ and $cx^2 + ax + b$ are positive for each $x \in \mathbb{R}$ and let $\alpha = \frac{bc + ca + ab}{a^2 + b^2 + c^2}$, then
 (A) $\alpha < 4$ (B) $\alpha < 1$ (C) $\alpha > \frac{1}{4}$ (D) $\alpha > 1$
-

Space for Rough Work

3. Let $f(x) = \operatorname{sgn}(\cot^{-1}x) + \tan\left(\frac{\pi}{2}[x]\right)$ (where $[]$ denotes GIF), then which of the following alternatives is/are true
- (A) $f(x)$ is many one but not an even function (B) $f(x)$ is periodic
 (C) $f(x)$ is bounded (D) graph of $f(x)$ remains above x-axis
4. If $f(x) = x^2, -1 \leq x \leq 4, g(x) = \sec^{-1}x, x \geq 1$, then
- (A) Domain of $\operatorname{gof}(x)$ is $[1, 4] \cup \{-1\}$ (B) Domain of $\operatorname{gof}(x)$ is $[-1, 4]$
 (C) Range of $\operatorname{gof}(x)$ is $[0, \sec^{-1}16]$ (D) Range of $\operatorname{fog}(x)$ is $\left(0, \frac{\pi^2}{4}\right)$
5. The last digit of $3^{81^n} + 1, n \in \mathbb{N}$, is
- (A) 4C_3 (B) 8C_7 (C) 8 (D) 4
6. m points on one straight line are joined to n points on another parallel straight line. The number of points of intersection of the line segments thus formed is : (excluding the given points)
- (A) ${}^mC_2 \cdot {}^nC_2$ (B) $\frac{mn(m-1)(n-1)}{4}$
 (C) $\frac{{}^mC_2 \cdot {}^nC_2}{2}$ (D) ${}^mC_2 + {}^nC_2$

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SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

-
1. Let $a_r = r^2 \frac{{}^{50}C_r}{{}^{50}C_{r-1}}$ and b denotes the coefficient of x^{49} in $(x - a_1)(x - a_2) \dots (x - a_{50})$, find $\left| \frac{b}{100} \right|$
 2. A variable line $\ell x + my = 1$ (ℓ, m are parameters) intersect a circle $x^2 + y^2 - 4x + 3 = 0$ at the point P and Q. The chord PQ subtends a right angle at the origin. If locus of the foot of perpendicular drawn from origin on the given variable line is $\lambda x^2 + \lambda y^2 - 4x + 3 = 0$ then find the value of $\frac{25}{4}\lambda$.
 3. The sum of squares of length of all possible chords intercepted by $x + y = n$, $n \in \mathbb{N}$ on the circle $x^2 + y^2 = 4$ is
-

Space for Rough Work

4. $S_n = \sum_{r=1}^n \frac{r}{1.3.5.7 \dots (2r+1)}$ then find the value of $5S_\infty$
5. If the solution set of the inequality $(\log_2 x)^4 - \left(\log_{1/2} \frac{x^3}{8}\right)^2 + 9\log_2 \left(\frac{32}{x^2}\right) < 4(\log_{1/2} x)^2$ is $(a, b) \cup (c, d)$, then find the value of $(2a + 3b + c + d)$. (where $a < b < c < d$)
6. Let $p_n = \cos\left(\frac{2^n \pi}{65}\right)$ and $q_n = \cos\left(\frac{(4n-2)\pi}{65}\right)$, where $n \in \mathbb{N}$. If $S = \left(\frac{\sum_{n=1}^{16} q_n}{\prod_{n=1}^5 p_n}\right)$ then find the value of S.

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SECTION-III : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

1. If $x + y + z = 10$, where $x, y, z \in \mathbb{N}$. Then the number of unordered triplets (x, y, z) satisfying the equation such that no two variables are equal _____
2. If $a, b, c \in \mathbb{R}^+$ and $a^2bc + 4abc + 4bc = 81$. Find the value of $a + b + c$ such that $2a + b + c + 4$ assumes its least value
3. Let $1 + \sum_{r=1}^{10} (3^r \cdot {}^{10}C_r + r \cdot {}^{10}C_r) = 2^{10} (\alpha \cdot 4^5 + \beta)$ where $\alpha, \beta \in \mathbb{N}$ and $f(x) = x^2 - 2x - k^2 + 1$. If α, β lies between the roots of $f(x) = 0$, then find the smallest positive integral value of k .
4. The number of three digit numbers of the form xyz with $x < y, z < y, x \neq 0$ is α then find the value of $\frac{\alpha}{60}$.
5. If $\cos\left(\frac{1}{4}\tan^{-1}\frac{24}{7}\right) = \frac{a}{\sqrt{b}}$ (where a and b are coprime natural number) then find $b - a$.
6. The number of real solution of the equation $x^3 + 1 = 2\sqrt[3]{2x - 1}$, is

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NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

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